MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination October 2025
Third Semester
MG3DSCFSQ201- Food Chemistry
(2024 Admission Onwards)

Duration: 2 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
1 mark each

1. b)The disruption of molecular order with starch granules
2. b) Lactose
3. c) polyphenolase
4. b) on the right
5. a) Heat
6. c) Acidic
7. d) None of these
8. b) Phospholipid
9. a) Decreases iodine value
10. b) Hydrolytic rancidity

Part B
Short Answer Questions
5 marks each

11. Write a note on the classification of monosaccharides.

Monosaccharides are the simplest carbohydrates that cannot be hydrolyzed into smaller sugar units. They are classified based on two main criteria — **functional group** and **number of carbon atoms**.

- Based on Functional Group
- Based on Number of Carbon Atoms
- Combined Nomenclature

(Explain each point briefly)

12. Define the terms: a) Water holding capacity, b) Water binding capacity, c) Water activity.
    - **Water holding capacity (WHC):** Ability of food components, especially proteins, to hold water under mechanical forces such as cutting or pressing.
    - **Water binding capacity (WBC):** The amount of water a material can bind per gram of dry material without being released during processing.
    - **Water activity (aw):** The ratio of vapor pressure of water in food to that of pure water at the same temperature; indicates availability of free water for microbial growth (range 0–1).

    (Explain each point briefly)

13. Write the structures of amino acids with R groups as aliphatic hydrocarbons.

Amino acids are the basic structural units (building blocks) of proteins.
General Structure : $NH_2-CH(R)-COOH$

Here, **NH_2** → amino group
COOH → carboxyl group
R (side chain) → variable group that determines the type and properties of each amino acid

- Aliphatic hydrocarbons are **organic compounds made of carbon and hydrogen atoms arranged in straight or branched chains** (not in rings).
- They can be:
    - **Alkanes (single bonds)** – e.g., methane, ethane
    - **Alkenes (double bonds)** – e.g., ethene
    - **Alkynes (triple bonds)** – e.g., ethyne

In the context of amino acids, an **aliphatic R group** means the side chain contains a straight or branched carbon chain without aromatic rings.

Amino Acid	R group structure	Characteristics
Glycine	H	Simplest, non-polar
Alanine	CH_3	Non-polar
Valine	$CH(CH_3)_2$	Branched, hydrophobic
Leucine	$CH_2CH(CH_3)_2$	Branched, hydrophobic
Isoleucine	$CH(CH_3)CH_2CH_3$	Branched, hydrophobic

14. Discuss the determination of Michaelis constant (Km) and Vmax.

The **Michaelis–Menten equation** describes the rate of enzyme-catalyzed reactions:

$$V = V_{max}[S] / K_m + [S]$$

Significance of Km and Vmax

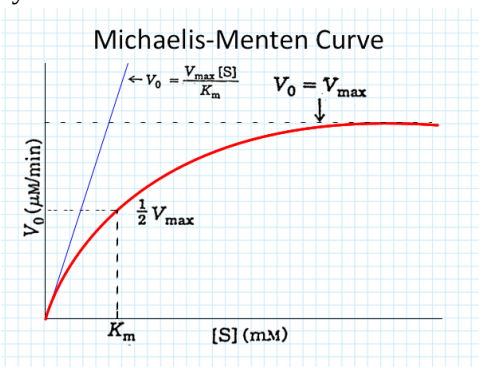
- **Km** measures the **affinity** of the enzyme for its substrate:
    - Low Km → High affinity (enzyme binds substrate tightly)
    - High Km → Low affinity
- **Vmax** represents the **maximum rate** achieved by the system when all enzyme active sites are occupied.

Experimental Determination

To determine **Km** and **Vmax**, the reaction rate is measured at various substrate concentrations.

- Prepare several reaction mixtures with constant enzyme concentration but varying substrate concentrations.
- Measure the initial reaction velocity (V_0) for each substrate concentration.
- Plot the data to determine Km and Vmax.

Graphical Methods- explain graph briefly



15. Explain how RM value helps in detecting adulteration in butter.

- **RM value** stands for **Reichert–Meissl value**.
- It measures the **amount of volatile, water-soluble fatty acids** (mainly butyric and caproic acids) in fat.
- It is expressed as the **millilitres of 0.1 N KOH** required to neutralize these acids from **5 g of fat**.
- **Pure butter fat** has a **high RM value** (around **24–32**).
- This is because butter contains **short-chain fatty acids** like **butyric acid**, which are volatile.
- **Vegetable oils and animal body fats** have **very low RM values** because they lack these volatile acids.
- When butter is **adulterated** with cheaper fats or oils, the RM value **decreases**.

- Therefore, a **low RM value indicates adulteration**, while a **normal value shows purity**.
- RM value is an **important quality control test** used in food analysis.
- It helps ensure **consumer safety and authenticity** of butter and ghee products.

16. Describe the difference between natural and synthetic antioxidants.

Type	Definition	Examples	Characteristics / Differences
Natural Antioxidants	Naturally occurring substances in foods that prevent oxidation by neutralizing free radicals.	Vitamin E (tocopherol), Vitamin C (ascorbic acid), β-carotene, Polyphenols, Flavonoids	• Found naturally in fruits, vegetables, oils, and spices. • Safe and health-promoting • Provide both nutritional and preservative benefits. • Usually less heat stable. • No harmful side effects.
Synthetic Antioxidants	Man-made chemical compounds added to foods to prevent oxidation and rancidity.	BHA (Butylated hydroxyanisole), BHT (Butylated hydroxytoluene), TBHQ (Tertiary butylhydroquinone), Propyl gallate	• Prepared artificially in laboratories. • More heat stable and effective in small quantities. • Used mainly as food preservatives. • May have regulatory limits for safe use. • Lack nutritional value.

Part C

Essay Questions

10 marks each

17. Explain the structure, composition, properties and industrial uses of starch and pectin.

Structure and Composition of Starch

- Starch is a **polysaccharide** made up of glucose units linked by **α-glycosidic bonds**.
- It occurs naturally in **plant storage organs** like cereals, potatoes, and roots.
- It has **two main components**:
- **Amylose (20–30%)** – linear chain of glucose molecules.
- **Amylopectin (70–80%)** – branched chain of glucose units.
- The **ratio of amylose and amylopectin** affects starch properties like gelatinization and viscosity.
- Starch granules are **insoluble in cold water** but swell and form a gel when heated.

. Properties of Starch

- **Gelatinization:** When starch is heated in water, granules absorb water, swell, and lose structure to form a thick paste.
- **Retrogradation:** On cooling, gelatinized starch may form a gel or recrystallize, leading to firmness (as in stale bread).
- **Hydrolysis:** Starch can be broken down by enzymes (amylases) or acids to produce dextrins, maltose, and glucose.
- Starch acts as a **thickening, gelling, and stabilizing agent** in many foods.

Structure and Composition of Pectin

- Pectin is a **complex polysaccharide** mainly composed of **galacturonic acid units** found in the **cell walls of fruits**.
- It can form **gels in the presence of sugar and acid**, which makes it important in jam and jelly making.

Industrial Uses

Starch Uses:

- Used as a **thickener** in soups, sauces, and gravies.
- Acts as a **binder and filler** in tablets (pharmaceutical industry).
- Used in **textile and paper industries** for sizing and coating.

Pectin Uses:

- Used as a **gelling agent** in jams, jellies, and marmalades.
- Used in **low-calorie foods** as a fat replacer.
- Acts as a **stabilizer** in fruit juices and dairy products.

18. Elaborate the structural classification of carbohydrates with examples.
    - Definition
    - Explain each types of carbohydrate with structure.

Type	Number of Units	Example	Function
Monosaccharide	1	Glucose, Fructose	Quick energy source
Disaccharide	2	Sucrose, Lactose	Energy supply
Oligosaccharide	3–10	Raffinose	Cell recognition
Polysaccharide	>10	Starch, Cellulose	Storage and structure

19. Explain in detail the structure of proteins.
    - Proteins are large biomolecules made up of **amino acids** linked by **peptide bonds**.
    - Their structure is organized at **four different levels** — primary, secondary, tertiary, and quaternary — each giving specific properties and functions to the protein.

Level of Structure	Description	Type of Bonds / Interactions	Example / Diagrammatic Meaning
1. Primary Structure	The **linear sequence of amino acids** in a polypeptide chain. Determines the protein's identity and function.	**Peptide bonds** (covalent) between the amino group of one amino acid and the carboxyl group of another.	Example: Ala–Gly–Ser–Val. A simple chain showing amino acids connected end to end.
2. Secondary Structure	The **regular folding or coiling** of the polypeptide backbone due to hydrogen bonding between nearby amino acids.	**Hydrogen bonds** between carbonyl oxygen and amide hydrogen atoms.	Two common types: • **α-Helix:** spiral structure (as in keratin) • **β-Pleated sheet:** zig-zag sheet (as in silk fibroin)
3. Tertiary Structure	The **three-dimensional (3D) folding** of the entire polypeptide chain into a specific shape. This determines the protein's **biological activity**.	Stabilized by: • Hydrogen bonds • Ionic bonds • Disulfide bridges (–S–S–) • Hydrophobic interactions	Example: Enzymes such as lysozyme or myoglobin. Globular structure with hydrophobic core and hydrophilic surface.
4. Quaternary Structure	The **association of two or more polypeptide chains (subunits)** to form a functional protein molecule.	**Non-covalent interactions** and sometimes disulfide bonds between subunits.	Example: **Hemoglobin** (4 subunits – 2α and 2β chains). Provides cooperative function.

20. Describe the classification of fatty acids.

Basis of Classification	Type	Description / Characteristics	Examples
1. Based on Saturation	**a) Saturated Fatty Acids (SFA)**	- Contain **no double bonds** between carbon atoms.- Solid at room temperature.- Usually found in animal fats and some plant oils (e.g., coconut oil).	Palmitic acid (C16:0), Stearic acid (C18:0), Myristic acid (C14:0)
	b) Unsaturated Fatty Acids (UFA)	- Contain **one or more double bonds** in the carbon chain.- Usually liquid at room temperature (oils).	- **Monounsaturated (MUFA):** One double bond → Oleic acid (C18:1)- **Polyunsaturated (PUFA):** Two or more double bonds → Linoleic acid (C18:2), Linolenic acid (C18:3)
2. Based on Chain Length	**a) Short-chain**	2–6 carbon atoms; water-soluble and volatile.	Butyric acid (C4:0)
	b) Medium-chain	8–12 carbon atoms; easily digested and absorbed.	Caprylic acid (C8:0), Lauric acid (C12:0)
	c) Long-chain	14–20 carbon atoms; most common in diet.	Palmitic acid (C16:0), Oleic acid (C18:1)
	d) Very long-chain	More than 20 carbons; found in waxes and sphingolipids.	Lignoceric acid (C24:0)
3. Based on Essentiality	**a) Essential Fatty Acids (EFA)**	Cannot be synthesized by the body; must be obtained from diet.	Linoleic acid (Ω-6), α-Linolenic acid (Ω-3)
	b) Non-Essential Fatty Acids	Can be synthesized in the body from other nutrients.	Palmitic acid, Oleic acid
4. Based on Configuration of Double Bonds	**a) Cis Fatty Acids**	Hydrogen atoms are on the **same side** of the double bond → causes a bent structure; natural form.	Oleic acid
	b) Trans Fatty Acids	Hydrogen atoms are on **opposite sides** of the double bond → straight structure; mostly from hydrogenation.	Elaidic acid (from margarine)